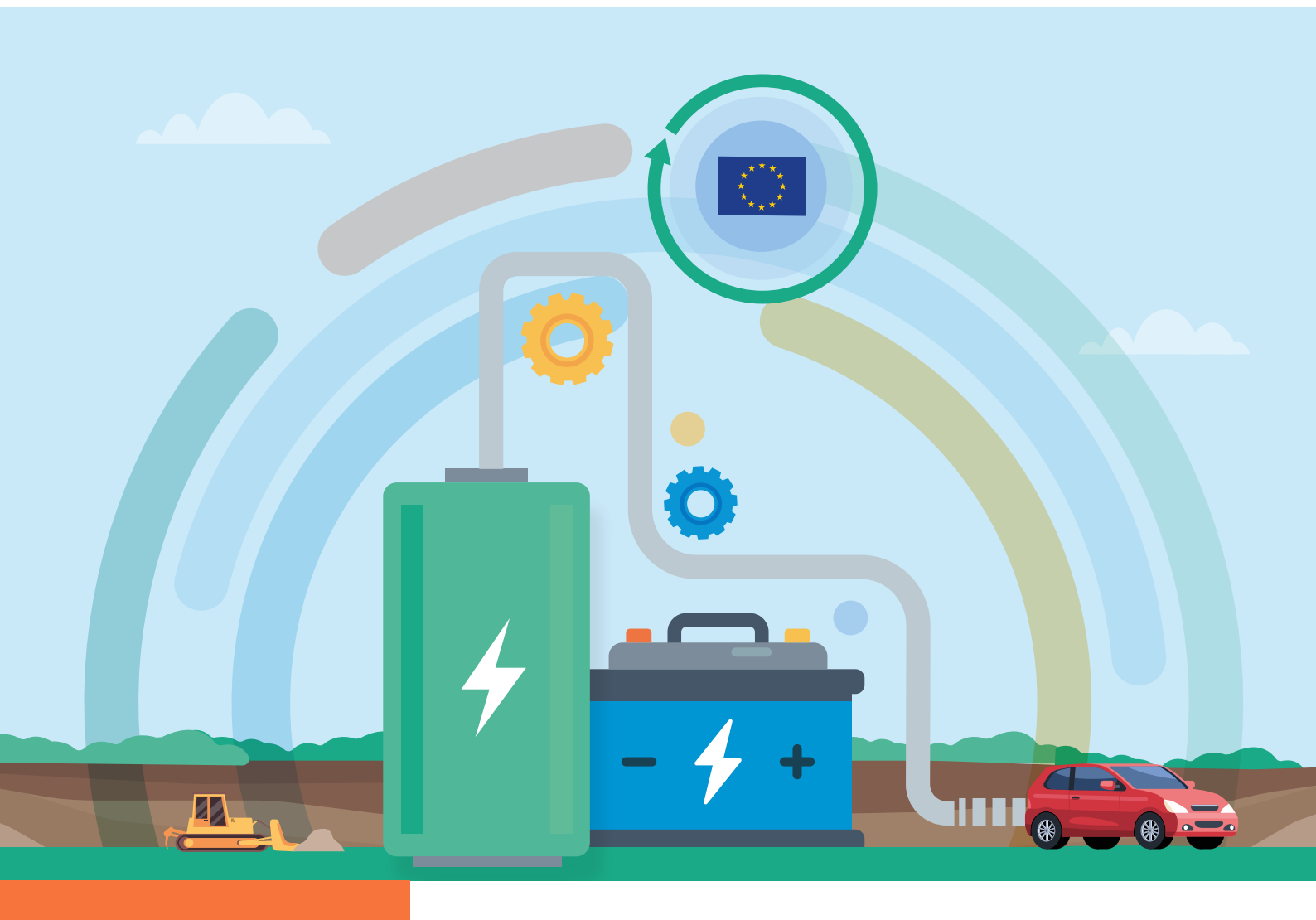


EU Battery Regulation: How to ensure it closes the circularity loop



About ECOS

ECOS - Environmental Coalition on Standards is an international NGO with a network of members and experts advocating for environmentally friendly technical standards, policies and laws. We ensure the environmental voice is heard when they are developed and drive change by providing expertise to policymakers and industry players, leading to the implementation of strong environmental principles.

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Executive summary

As Europe decarbonises, batteries (and the materials needed to manufacture them) are in high demand. Used for many different purposes – including electric vehicles (EVs), lighting, and industry – batteries must be sustainable and safe while addressing growing environmental concerns that stem from widespread electrification. As well as reducing the carbon footprint of manufacturing, the right solutions will also account for the end of a battery's life.

For batteries to be sustainable, their lifespan must be prolonged, and their lifecycle impacts properly managed – from the extraction of raw materials and manufacturing to repair and waste disposal. Batteries need to be durable, repairable, reusable, and recyclable. They also need to use materials more efficiently, including by decreasing their overall use, using them for longer, and incorporating more recycled materials into production – closing the circularity loop.

Focusing on multiple types of batteries and their components¹, the EU Battery Regulation² tries to achieve this using a variety of tools – including secondary legislation (summarised in Annex I), harmonised standards³, labelling, and information sharing. It will come to life in phases, from February 2024 until 2036, and apply to all types of batteries on the EU market – whether produced in the EU or imported. The regulation applies to batteries that are incorporated into products or designed for a specific use.

This report evaluates the EU Battery Regulation's effectiveness, focusing on electric vehicle (EV) batteries – one of the most widespread and essential uses for batteries in Europe, due to the urgent need to decarbonise transport. We evaluate whether the legislation addresses circularity for batteries in a successful way, focusing on the expected Delegated and Implementing Acts and the requested standards (as of March 2024).

An ambitious piece of legislation

Overall, the EU Battery Regulation is an ambitious piece of legislation that promotes sustainable practices in the battery industry. Adopted in 2023 to replace its outdated predecessor⁴, it must now be implemented appropriately – but this is not a given. To account for evolving environmental challenges and technological advancements, collaboration among industry stakeholders to refine this legislation will be crucial to its success.

The regulation emphasises a full lifecycle approach, targeting battery components⁵ from design and manufacturing to use and end-of-life. It establishes requirements for documenting carbon footprint, recycled content, performance and durability, safety, state of health (SoH) information, battery repurposing, and recycling efficiencies. It also introduces a digital battery passport, due diligence and risk management policies for companies, as well as mandatory third-party verification. This will enhance accountability, traceability, and transparency in the battery value chain.

This progressive piece of legislation will help to close the circularity loop for batteries, establishing a new framework and systems that are underpinned by principles of increased efficiency and waste reduction – with impacts throughout global supply chains.

However, with secondary legislation and standards still under development, the regulation cannot yet be fully evaluated – and it contains some potential weaknesses. For example, explicit design requirements are omitted, and insufficient attention is paid to the mining and

processing stages for battery materials. We are also concerned by carbon footprint calculations, which are yet to be defined by the European Commission in an upcoming Impact Assessment. Implementation challenges are also possible due to watered-down standards.

During the implementation phase of the EU Batteries Regulation, we must pay close attention to all the potential issues to ensure that a truly circular economy for batteries is realised.

Recommendations

To ensure that a truly circular economy for batteries is realised, during the implementation phase of the EU Battery Regulation, ECOS recommends the EU and its Member States to:

1

Prioritise sufficiency by reducing the need for primary raw materials (that mostly do not originate from Europe). Mining and processing impacts should be addressed in EU legislation⁶,

4

Closely monitor state of health-related provisions to ensure they lead to a flourishing second-hand market for batteries. They must not undermine its development due to a lack of definitions or guidelines.

2

Enhance resource efficiency by reducing the size of EV batteries (and, subsequently, vehicles), while promoting sustainable design. This must be reflected in carbon footprint calculations and any relevant thresholds.

5

Raise the recycling efficiency for lithium to reflect the need for this material as an essential component of EV batteries.

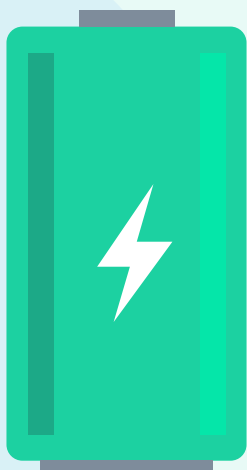
3

Introduce additional provisions to effectively address the design of battery components.

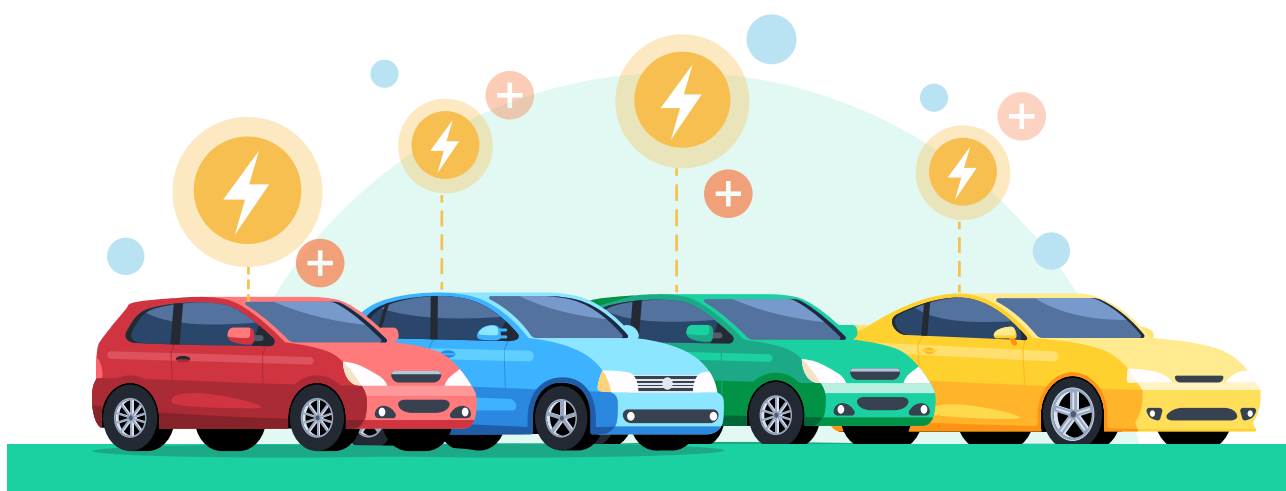
Parts must be removable, replaceable, repairable, and recyclable.

6

Harmonise performance and durability provisions with existing technical regulations, giving priority to increasing circularity through increased battery lifespans, as per one of the aims of the Battery Regulation.



Circularity and the EU Battery Regulation



Batteries are resource hungry products that need to be reimagined – from design to end of life – to align with the need for a transition to clean energy and a circular economy.

Increased demand for batteries due to the rapid electrification of vehicles inevitably leads to increased mining activity, which impacts the environment and local communities. Therefore, as a first step, making batteries smaller (particularly for EVs) is a logical way to reduce material use. This, in turn, would require smaller vehicles, so would ultimately save resources in the manufacturing stage. Smaller batteries also imply a lower carbon footprint, which is needed to meet decarbonisation targets and make the low-carbon economy a reality.

Another important aspect of battery circularity is the length of their life. As well as ensuring that batteries can last for as long as possible during their first life, they should be given a second life through repair, reuse, remanufacturing, and repurposing. Such steps must be supported by robust

policy frameworks and standards that facilitate the creation of a second-hand market. These activities can be supported by easier access to information on a battery's "identity" and "state of health" to allow those working in the second-life stage to base their choices on accurate data. Information on a battery's profile should also be transparently available to end-users, to empower them to make informed choices. Such steps would help to discourage the use of batteries with a high environmental impact.

To close the circularity loop, batteries must be collected and recycled at their definite end of life, and the resulting inevitable waste should be managed adequately. High-quality recycling is particularly important for battery manufacturing with a larger percentage of recycled materials. Such recycling is energy-intensive, so it should be limited to the end of life. Access to secondary raw materials is vital for the EU's raw materials sovereignty, as well as helping to reduce the need for extraction and processing.

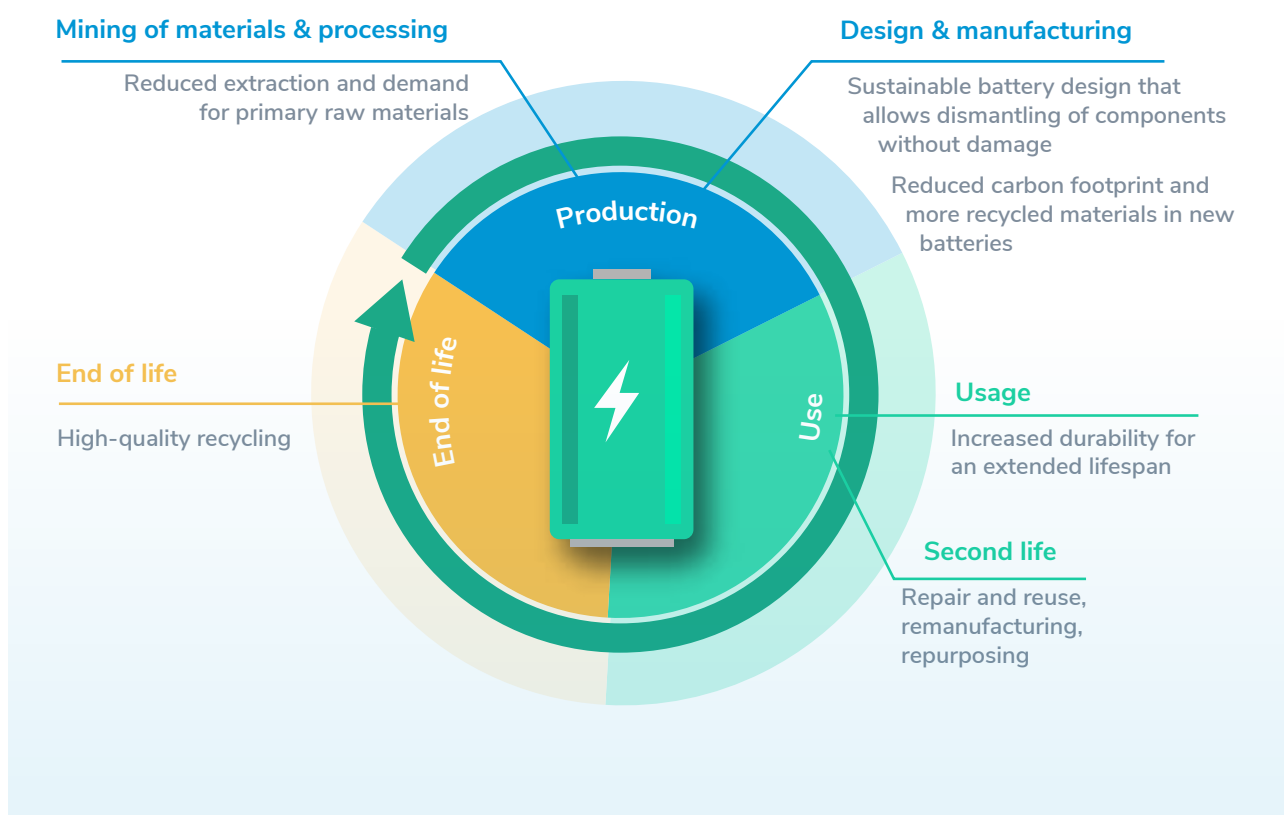


Figure 1 Representation of circularity loop for batteries.

The EU Battery Regulation lists a number of provisions of relevance to the circularity of batteries, including:

Articles 9 and 10

outline criteria for battery performance and durability, while Article 11 covers the removability and replaceability of portable batteries in appliances, promoting sustained use even if batteries have a shorter lifespan.

Article 73

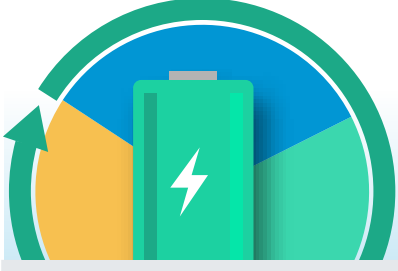
encourages the repurposing and remanufacturing of batteries, and Article 74 compels producers to provide information facilitating battery dismantling and removal, supporting repair, remanufacture, preparation for reuse, treatment, and recycling.

Articles 8, 59, 69, and 71

contribute to closing material cycles by setting requirements for recycled content and efficiencies. More specifically, Article 8 dictates that cobalt, lead, lithium, and nickel in new electric vehicle batteries must meet minimum requirements for material recovery from waste for each metal.

The regulation also requires the implementation of a battery passport containing details about repair, disassembly, and, notably the carbon footprint of the battery, spanning from raw material extraction to the end-of-life phase, excluding the usage period. Manufacturers must also report information on the state of health (SoH) of batteries – information that is essential for their repair and second life.

The Regulation also establishes baseline criteria for Extended Producer Responsibility and due diligence requirements for the battery supply chain to help combat environmental and human rights violations.



Battery circularity elements	Stage	In EU Battery Regulation
 Mining and processing	Production	 No/not in scope
 Design and manufacturing of batteries	Production	 Yes
 Usage of batteries (extended lifespan)	Use	 Yes
 Second life	Use	 Yes
 Recycling	End of life	 Yes
 Waste management	End of life	 Yes

Figure 2 What circularity boxes does the EU Battery Regulation tick?

What does the EU Battery Regulation overlook?

Although the Battery Regulation encompasses most of the stages required to develop a good circularity framework for EV batteries in the EU, one is missing. The regulation does not address the mining and processing stages for materials used in batteries (apart from the due diligence requirements and the calculation of the carbon footprint of the batteries).

Mining and processing may well be out of the scope of this piece of legislation. However, no reference is made to another relevant law or to the need for mining within environmental boundaries (to the extent that it is possible). The Battery Regulation creates no link between reduced raw material extraction through battery design, nor applications that favour sufficiency and reduced material use as well as efficiency.

The circularity loop can only be considered complete if one overlooks that the regulation omits the very beginning of a battery's life, taking for granted that increased mining activities will be addressed through recycling and due

diligence policies – which is not the case. In line with the need to address the mining and processing stages, ECOS supports⁷ the adoption of an EU Sustainable Resource Management policy framework to fill the legislative gap on sufficiency strategies to reduce resource overconsumption.

Another weak point of the Battery Regulation is that it does not discuss battery design explicitly, undermining the important role that production plays in closing the circularity loop. Although performance and durability requirements are set in the Battery Regulation (and in the Euro 7⁸), there are no provisions for minimum design requirements concerning the assembly and disassembly of an EV battery parts. Performance and durability are only one piece of the puzzle – to design a battery and its components in a sustainable way, other things also need to be considered. How a product is designed significantly affects how it can be repaired and recycled. Parts must be safely accessible, removable, and replaceable to ensure high circularity for EV batteries.



Provisions for EV batteries in the EU Battery Regulation

The EU Battery Regulation sets provisions that will help to create a circular value chain and end-of-life treatment for EV batteries. For example, it introduces requirements for carbon footprint calculation, performance and durability, recycling content and efficiencies, and a battery passport,

There are also certain provisions on repair and repurposing as well as reporting on the state of health of EV batteries. How robust are each of these provisions? How could they be further improved? What gaps will need to be addressed in the future? We provide answers below.

Battery Regulation provisions	Evaluation
 Carbon footprint	 Good but with certain issues
 Performance and durability	 Good but with space for improvement
 State of health	 Challenged
 Battery passport	 Good but with space for improvement
 Repair, repurposing, remanufacturing	 Challenged
 Recycled content	 Good but with space for improvement
 Recycling efficiencies	 Good except lithium target
 Minimum design requirements	 Missing

Figure 3 Evaluation of the EU Battery Regulation



Carbon footprint

Article 7 and Annex I

In the EU Battery Regulation, determining the carbon footprint of batteries follows the European Commission's product environmental footprint (PEF) method and the applicable Product Environmental Footprint Category Rules (PEFCRs), such as that for high specific energy rechargeable batteries designed for mobile applications. The battery's carbon footprint should also align with international agreements and lifecycle assessment (LCA) technical guidance, including international standards such as ISO 14040:2006 and ISO 14067, which specifically addresses carbon footprint considerations.

Article 7 and Annex I introduce the requirement for a carbon footprint declaration for batteries through a Delegated Act (18 February 2025) that includes a calculation methodology, and their eventual classification in different footprint classes as a second step. The aim of this gradual approach is to set a maximum carbon threshold for batteries as a mandatory condition for their placement on the EU market.

For the carbon footprint declaration, the regulation includes guidelines on the lifecycle stages to be reported, such as raw material acquisition and processing, production, distribution, end-of-life, and recycling. Use is excluded from the carbon footprint calculation. In addition, specific information should be provided on the battery model as well as the manufacturing plant. The carbon footprint of the battery will be contained as information on its digital passport.

ECOS analysis

For the carbon footprint calculation, valid and up-to-date data is needed from the battery supply chain. It is unclear whether manufacturers are always able to include primary (instead of secondary) data in their LCA. This could lead to a potential underestimation of the environmental impact of certain batteries.

A good calculation for carbon footprint that reflects reality is the only way to make decarbonisation possible through the electrification of vehicles. A carbon footprint threshold is needed to incentivise smaller size batteries with less impact.

As the carbon footprint calculation will be set by the Delegated Act (which has not yet been published), we cannot yet evaluate the quality of the calculation – only the calculation suggested by the Joint Research Centre (JRC) in their report⁹. We make this analysis in the section below on secondary legislation.



Performance and durability

Article 10 and Annex IV

From August 2024, electric vehicle batteries must be accompanied by documentation on their electrochemical performance and durability. Performance criteria encompass aspects such as rated capacity, power, internal resistance, and energy round trip efficiency. Durability criteria include capacity fade, power fade, internal resistance increase, and round-trip efficiency fade. While rated capacity, power, and internal resistance are contingent on battery size, all other parameters are normalised quantities.

Minimum values on electrochemical performance and durability parameters are to be set through a Delegated Act, which is scheduled to be published by 18 August 2027.

The Standardisation Request for batteries mandates standards that will describe the necessary steps and conditions for the measurement of parameters that are relevant for the initial application and (if intended for the battery) reuse and repurposing. These should be in place as of 7 June 2026.

ECOS analysis

As well as in the Battery Regulation, performance and durability are also being regulated under the UNECE regulation GTR no. 22 for Electric Vehicles¹⁰, with certain provisions also in the Euro 7 legislation. As this is already regulated in many places, provisions must be harmonised and not disregard the aim of the Battery Regulation to increase circularity through an extended lifetime.



Information on the state of health (SoH) and expected lifetime of batteries

Article 14

Article 14 outlines requirements related to information on the state of health and expected lifetime of batteries. Starting from 18 August 2024, the battery management system of EV batteries must contain up-to-date data based on specified parameters.

The information will be accessible in a read-only format and must be provided to those who legally purchase the

battery, allowing them to make the battery available for energy storage, assess its residual value or remaining lifetime, and facilitate preparation for re-use, repurposing, or remanufacturing. The European Commission has requested harmonised European standards to describe procedures for the determination of the SoH of batteries by June 2027.

ECOS analysis

Knowledge of a battery's SoH is necessary to make the second-hand market for batteries reliable and operational.

Battery manufacturers have voiced concerns about providing information on the SoH of batteries because, they claim, indiscriminate access to data could impact fair competition. Other concerns relate to the parameters taken into consideration when calculating the SoH of a battery and how they are affected by real-life applications beyond the theoretical approach.

The SoH information of batteries should be clearly defined in standards, and it should remain up to date throughout the first life of the battery¹¹. Pertinent information should also be available to interested parties, and repurposers should not be limited by the original battery manufacturer – repurposing should be able to take place in independent facilities, not only that of the original manufacturer.



Battery passport

Chapter IX and Annex XIII on information to be included

From February 2027, every EV battery should also have its own digital battery passport. This must include details on the battery model, basic information on the manufacturer such as name and address, declarations of conformity with EU legislation, references to relevant harmonised standards, and any acquired certification.

The battery passport must be fully interoperable with other digital product passports and accessible by interested parties free of charge. The right to access, introduce, modify, or update information in the battery passport shall be restricted based on the access rights of different categories and should be limited to economic operators and repurposers who prepare the battery for its second life.

The purposes for accessing and processing the information concern dismantling of the battery, including safety measures to be taken during this process, and the detailed composition of the battery model. This is essential to allow repairers, remanufacturers, second-life operators, and recyclers to access the information they need. As a result, it significantly supports circularity and facilitates the extension of life of EV batteries.

ECOS analysis

ECOS strongly supports the introduction of the battery passport. We also want to ensure that all information reported is accurate and up to date. Links must be more clearly defined between the technical documentation of performance and durability parameters and data on state of health and remaining lifetime.

Clarity is also needed on the way the usage data of batteries will stay updated and reflect real-life use (especially when it comes to state of health). While the manufacturer will be responsible for this, and will have editing access, no concrete plan has been presented to date.



Repair and remanufacturing or repurposing

Article 73

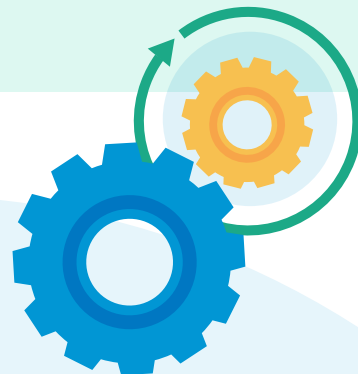
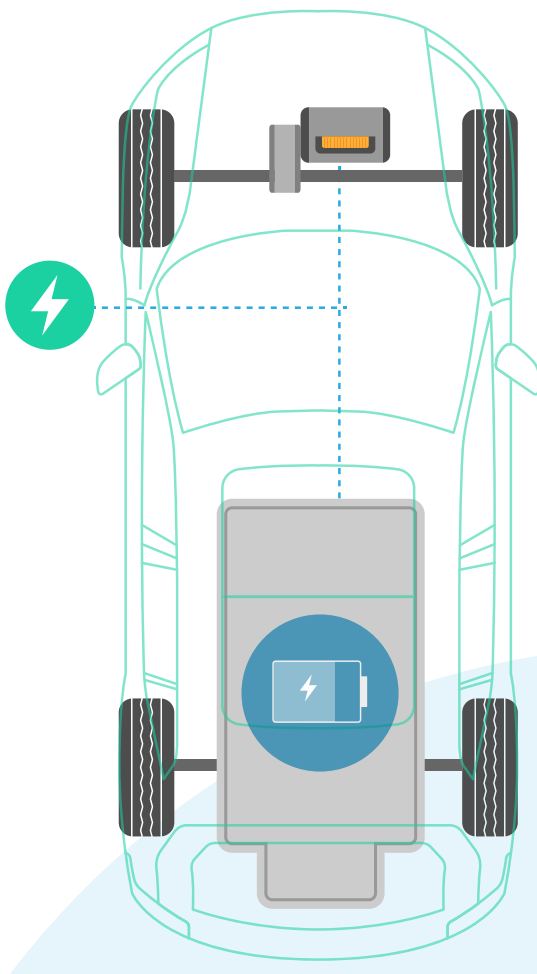
Article 73 describes the details that should be made available to prepare batteries for their second life. The most important piece of information is the SoH of the battery, which must be supported by evidence through evaluation or testing carried out within the EU.

The European Commission has requested the creation of standards to describe the process of evaluating batteries for repair or repurposing. The standard must provide guidance on the necessary steps, conditions, and protocols for the safe repair, re-use, and repurposing of batteries and battery packs, modules, and cells originally designed for electromobility applications.

ECOS analysis

Repair, repurposing, and remanufacturing are all important steps in the circularity loop for batteries. Such explicit recognition by the EU (in the form of a regulation) that batteries need to be given a second life, is a much-needed signal for car manufacturers, who must now become more supportive of the second-hand EV market.

However, as the regulation lacks minimum design requirements that will ensure the removability, repairability, and replaceability of EV batteries, it sets unstable foundations for effective implementation and the encouragement of second-life applications. Standards will be the only place that repair and repurposing will be discussed in detail – and only with guidance, not binding measures – which creates uncertainty about how progressive the industry's approach will be on these issues.





Recycled content and recycling efficiency and recovery of materials

Articles 8 & 71 and Annex XII

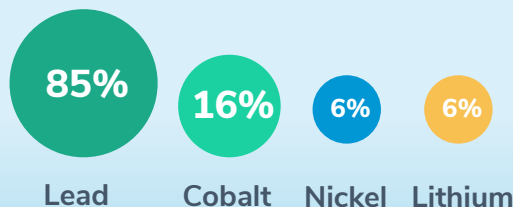
Article 8 sets an obligation for batteries that contain cobalt, lead, lithium, or nickel in active materials to be accompanied by documentation that mentions what percentage of those has been recovered from battery manufacturing or post-consumer waste. The provision is to go into effect with a Delegated Act on 18 August 2028. By August 2026, a Delegated Act should be adopted on the methodology for the calculation and verification of the percentage of recovered cobalt, lithium, or nickel used in new batteries, as well as the percentage of lead.

These obligations, however, will not apply to batteries placed on the market and put into service before the provisions go into effect, and which are being or have been repurposed or remanufactured. Targets will be reassessed in 2028 based on the availability or lack of the materials in scope and revised, if necessary.

Recovery rates for permitted facilities are set in Annex XII of the Battery Regulation, while a Delegated Act establishing a calculation and verification methodology for recycling efficiency and recovery of materials should be adopted by 18 February 2025. The first targets for recycling efficiency should be reached by 31 December 2025 and increased by the end of 2030. Recycling efficiencies will reach 80% by average weight of lead-acid batteries, 70% of lithium-based batteries, 80% for nickel-cadmium batteries, and 50% for other waste batteries.

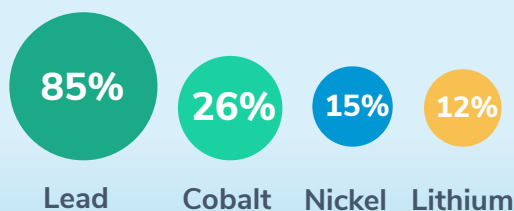
Starting from August 2031

the Battery Regulation requires batteries placed on the European market to contain the following percentages of recovered materials:



In 2036

the requirements for recovered materials will increase to:



ECOS analysis

A weak point of this article is that recycled content will not only come from post-consumer waste – as we had originally advocated – but will also include post-industrial waste, such as metal scraps, in a higher percentage than preferable. By prioritising post-consumer waste over post-industrial waste, market value would have been given to materials that otherwise would be the least desirable for manufacturers.

For lithium-based batteries – an essential component of EV batteries – we had advocated for the percentage of recycling efficiency to be set at 90%. As this was only set at 70%, this element could have been stronger.

Developing secondary legislation and standards

Discussions on the required Delegated Acts have started and are progressing at pace, with a publication deadline of February 2025. A draft text is expected to be publicly available for consultation in April 2024.

While exchanges on the content were not always accessible to NGOs, ECOS has contributed to discussions on the Delegated Acts for the carbon footprint calculation methodology, recycling efficiencies, and recycling content. We have also provided feedback on the performance and durability of EV batteries.

With the process ongoing, the most important point to underscore is that the carbon footprint methodology developed by the European Commission should be future proof. It must not encourage the production of batteries with an unnecessarily large carbon footprint – such as those used for larger electric cars (like SUVs).

According to the most recent JRC draft report¹², the carbon footprint will be calculated by considering the 'energy provided over the service life' of a battery. This calculation will be based on information about the battery's durability and the general energy consumption of the vehicle in question, which will also determine the carbon footprint performance class of the battery. The energy consumption of vehicles per kilometre is determined through the Worldwide Harmonized Light Vehicles Test Procedure (WLTP)¹³. Different types of vehicles have different energy consumption. As a result, the type of vehicle will influence the calculation and the eventual carbon footprint classification of the battery. Because of the methodology

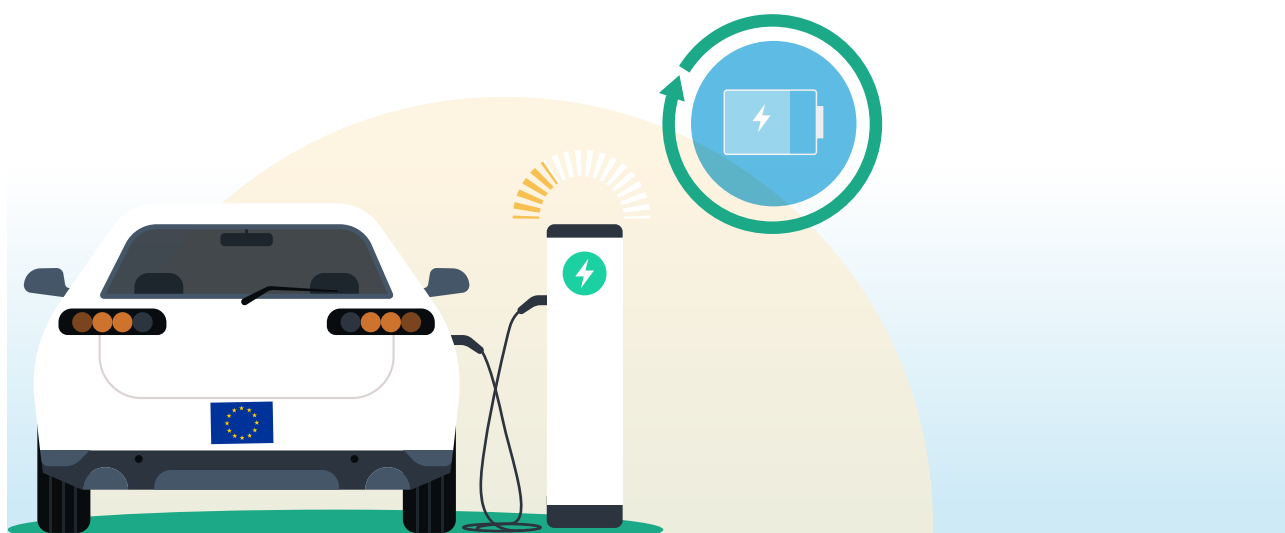
used for the calculation, a battery could end up receiving a better carbon footprint performance class than appropriate, because the vehicle it is used in has a higher energy consumption per km under WLTP.

To set the threshold limit for battery carbon footprint, this calculation must not favour high energy consuming, heavy, or energy inefficient vehicles. To avoid this, ECOS had proposed to the JRC that service life should be calculated by using the fixed energy consumption estimated for M1 vehicles – defined by the JRC based on market data – and that the absolute battery carbon footprint should also be made available for each battery put on the market to increase the amount of information available.

Regarding the standardisation process, ECOS is closely monitoring relevant activities. We recognise the hesitation of standardisation organisations to promote innovation and sustainability through controversial topics such as the state of health. However, given that the requested standards will be part of EU legislation and have a binding nature, they should be aligned with the environmental ambition of the legislator, and not side-tracked.

We also regret the slow process and significant delays in the creation of harmonised European standards, as requested by the European Commission. Although no provisions of the Battery Regulation seem to be affected, the almost two additional years offered for certain standards were not necessary, given that the initial Standardisation Request was published in 2021 and there was plenty of time to progress in the technical committee working groups.

Conclusion



The EU Battery Regulation has taken a brave step forward when it comes to circularity. The lifecycle of batteries has been addressed (mostly) adequately for the scope and it should be enhanced by other pieces of legislation in the future.

However, certain measures are better supported and more robust than others, while others are exposed to potential loopholes. For the moment, loopholes are mostly associated with a lack of prescriptive measures and stricter requirements. But there is a risk that others could arise, either through inadequate calculation methodologies or standards that demonstrate low ambition and desire to move towards a second-hand market for EV batteries.

Overall, the EU Battery Regulation, at least from a legislative point of view, is a big win. It will promote better practices

in the battery industry and is more progressive than other similar legislation in other parts of the world. However, the implementation phase may prove challenging. Certain industry players lack ambition and there are different realities in EU Member States when it comes to data collection, existing facilities for second-life activities, recycling, and waste management.

The text of the EU Battery Regulation has been agreed upon, but its full impact remains to be seen. With certain questions still hanging over the ongoing implementation process, now is not the time for complacency. There are still ways that this legislation could be more successful – or fail. Let's keep our foot on the accelerator and continue working to close the circularity loop for batteries in the EU.

Annex I

Secondary legislation underpinning implementation of the EU Battery Regulation

The table below lists the most relevant secondary legislation¹⁴ and standards requested for electric vehicle (EV) and portable batteries¹⁵:

Article	Secondary legislation	Standards	Deadline
Article 7 Carbon footprint technical documentation	Methodology to calculate the total carbon footprint of the battery through a Delegated Act, with a format to be established with an Implementing Act	Not requested	18 February 2025
Article 7 Carbon footprint performance classes	Carbon footprint performance classes to be established with a Delegated Act, and a format for the labelling established by an Implementing Act. + PEF category rules for batteries	Not requested	18 February 2025
Article 8 Recycled content for industrial and EV batteries	Methodology for the calculation and verification of the amount of cobalt, lead, lithium or nickel, and its format through an Implementing Act	Not requested	18 August 2028
Article 9 Performance and durability requirements for portable batteries	Minimum values for the electrochemical performance and durability parameters through a Delegated Act	Standards to describe the measurement methods necessary for the determination of the performance and durability of portable rechargeable and non-rechargeable batteries of general use	18 August 2027 (DA) 7 June 2026 (Standards)
Article 10 Performance and durability requirements for rechargeable industrial batteries and EV batteries	Minimum values to be established through a Delegated Act	The standard(s) shall describe the necessary steps and conditions for the measurement of the parameters, which are relevant for the initial application and (if intended for the battery) the reuse and repurposing application	18 February 2026 (DA) 7 June 2026 (Standards)

Article	Secondary legislation	Standards	Deadline
Article 12 Safety	Delegated Act to amend the safety parameters laid down in Annex V in view of technical and scientific progress	Standards on safety requirements	No deadline for DA 7 June 2026 (Standards)
Article 14 State of health	Not requested	Diagnostics and determination of the state of health (SoH): The standard(s) shall describe the procedure(s) for the determination of the SoH of batteries	7 June 2027
Article 59 Battery repurposing	Not requested	Battery evaluation for repairing or repurposing: The standard shall describe the necessary steps, conditions and protocols for the safe repair, re-use and repurpose of batteries and battery packs, modules and cells originally designed for electro-mobility applications	7 June 2027
Article 71 Recycling efficiencies and material recovery targets	Delegated Act to establish the methodology for calculation and verification of rates for recycling efficiency and recovery of materials, in accordance with Part A of Annex XII, and the format for the documentation	Not requested	18 February 2025

Notes and references

- 1 Including portable batteries, starting, lighting, and ignition batteries (SLI batteries), light means of transport batteries (LMT batteries), electric vehicle batteries, and industrial batteries.
- 2 EU Battery Regulation (Regulation 2023/1542): <https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32023R1542>
- 3 Standardisation request M/579: https://ec.europa.eu/growth/tools-databases/enorm/mandate/579_en
- 4 Directive 2006/66/EC: <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32006L0066>
- 5 Battery components, such as cathodes, anodes, separators, electrolytes, connectors, casings, safety equipment, and battery management systems contribute to environmental impacts throughout their supply chain, including the mining, processing, and fabrication stages.
- 6 Microsoft Word - EN Global Justice Letter to the EU_incl signatories (002).docx (eeb.org)
- 7 <https://ecostandard.org/publications/sustainable-resource-management-directive/>
- 8 Euro 7 is the revised EU regulation for the type-approval of motor vehicles and engines, and of the systems, components and separate technical units intended for such vehicles, with respect to their emissions and battery durability. It sets more adequate rules for vehicle emissions and aims to further lower air pollutant emissions from road transport, and for the first time it covers cars, vans, and heavy-duty vehicles in one single legal act. A provisional agreement was reached on Euro 7 in December 2023. The new law will go into effect within 2024 after its official approval by the European Parliament.
- 9 GRB-CBF_CarbonFootprintRules-EV_June_2023.pdf (europa.eu)
- 10 UN GTR No. 22 “In-vehicle Battery Durability for Electrified Vehicles” is a global technical regulation established by the United Nations Economic Commission for Europe (UNECE) to address the durability and performance of batteries used in electrified vehicles. Specifically, it focuses on battery health monitoring and related data availability for electric and plug-in hybrid vehicles.
- 11 <https://ecostandard.org/the-positive-side-of-batteries/>
- 12 GRB-CBF_CarbonFootprintRules-EV_June_2023.pdf (europa.eu)
- 13 The Worldwide Harmonized Light Vehicles Test Procedure (WLTP) is a global standard for determining the levels of pollutants, CO₂ emissions and fuel consumption of traditional and hybrid cars, and the range of fully electric vehicles.
- 14 The law also requests secondary legislation for the labelling of batteries, due diligence requirements, collection rates of portable and LMT batteries, and green public procurement.
- 15 The deadlines mentioned do not reflect all enforcement dates within the Battery Regulation, but those concerning the relevant Delegated Acts and standards. Additionally, standardisation request deadlines have been extended since the amendment process for M/579 began so are consequently longer than the initial deadline of 7 December 2025 for all requested standards.



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